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Smart Parking Management and Predictive Space Allocation Using Queues and Hash Tables

CSA0311-DATA STRUCTURES

GUIDED BY:

DR K.KIRUTHIKA

DONE BY:

RANJITHA R-192421285

NASINA UMASRI -192424240

KIRUTHIKA B-192511232



INTRODUCTION



- The increasing number of vehicles creates a growing demand for efficient parking management.
- Traditional parking systems can cause delays, congestion, and inefficient use of parking spaces.
- Our proposed Smart Parking Management System provides an automated way to manage parking slots.
- The system uses graphs, priority queues, hash tables, queues, and linked lists for efficient parking operations.
- It provides smart slot allocation, navigation, vehicle search, waiting-queue management, billing, and parking history.
- Predictive analytics helps estimate parking demand and improve parking space utilization.



OBJECTIVE



- To develop an efficient and automated system for managing available, occupied, and reserved parking spaces.
- To automatically allocate the most suitable parking slot based on availability and priority.
- To reduce vehicle waiting time, parking search time, and congestion within the parking area.
- To implement graphs, priority queues, hash tables, queues, and linked lists for efficient parking data management.
- To provide quick vehicle search, optimized navigation, billing, and parking history management.
- To analyze parking usage and predict future parking demand for better space utilization.

PROBLEM STATEMENT

- Traditional parking systems often lack real-time information about available and occupied parking slots.
- Drivers may spend more time searching for suitable parking spaces, leading to delays and congestion.
- Manual management of vehicle records, parking slots, and waiting queues can be inefficient.
- Available parking spaces may not be utilized effectively due to inefficient slot allocation.
- Many parking systems lack optimized navigation to guide vehicles to their allocated slots.
- Existing systems may not provide predictive analysis to estimate future parking demand and improve space utilization.



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LITERATURE SURVEY



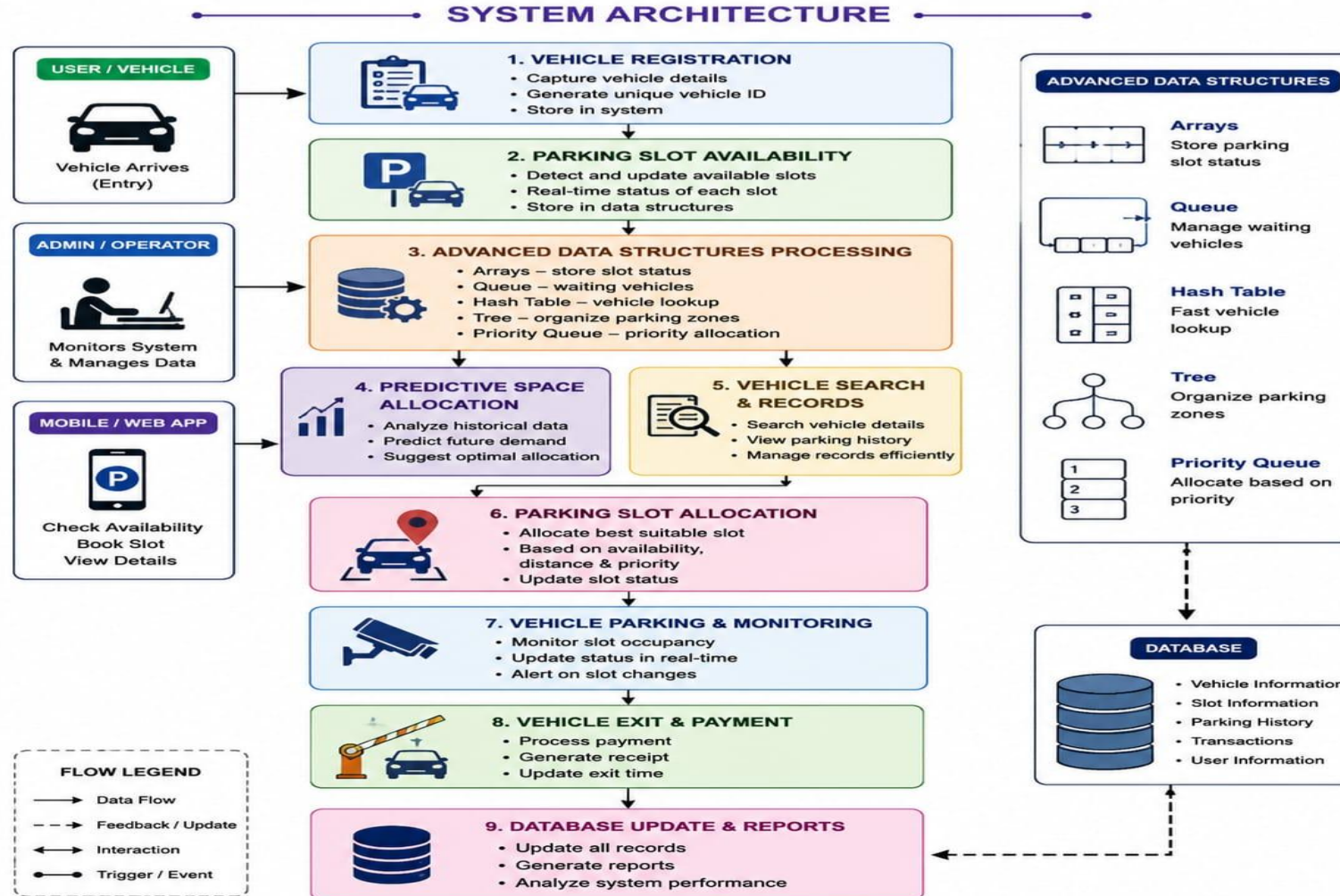
Journal Name	Author	Year	Key Finding	Limitation
Journal of Traffic & Transportation Engineering	Ma et al.	2024	Improved parking demand prediction	Requires quality data
Transportation Research Procedia	Channamallu et al.	2023	IoT and sensors enable real-time parking	High infrastructure cost
EPJ Data Science	Kaltenbrunner et al.	2025	Predicts parking occupancy	Limited to occupancy prediction
Green Energy & Intelligent Transportation	Channamallu et al.	2026	AI, IoT and real-time monitoring improve parking	Scalability and sensor issues



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ARCHITECTURE



MODULE 1: SMART PARKING REGISTRATION AND SLOT ALLOCATION

- Vehicle Registration: Register vehicle details such as vehicle number and owner information during entry.
- Unique Parking ID: Generate a unique parking ID for each registered vehicle for easy identification and tracking.
- Real-Time Availability: Check the current status of available, occupied, and reserved parking slots.
- Smart Slot Allocation: Automatically select and assign the most suitable available parking slot to the vehicle.
- Status & Data Management: Update slot status from Available → Occupied and store vehicle and parking details for future reference.

MODULE 1: OUTPUT

 **SmartPark AI**
CSA0311 – Data Structures

Dashboard

Vehicle Registration

Smart Slot Allocation

Parking Map & Navigation

Waiting Queue

Vehicle Search

Billing

Parking History

Predictive Analytics

Algorithm Comparison

Data Structures Visualizer

Admin / Simulation

Project Info

Vehicle Registration

Register New Vehicle

Vehicle Number

TN09AB1234

Owner Name

Owner name

Vehicle Type

Car

Special Requirement (optional)

None

Register & Auto-Allocate

Parking ID P1014 → Slot C5 allocated (score 79/100)

Last Registered Vehicle

Vehicle Number

TN76AP5437

Parking ID

P1014

Allocated Slot

C5 (Regular)

Smart Score

79/100

✓ Shortest estimated travel time

✓ Low congestion

✓ Ideal match for vehicle type

✓ Good predicted future availability

Advanced Data Structure-Based Smart



MODULE 2: INTELLIGENT NAVIGATION AND PARKING OPERATIONS



- **Nearest Slot Identification:** The system identifies and displays the nearest suitable available parking slot, helping drivers quickly locate an appropriate space.
- **Optimized Navigation:** It guides the vehicle from the parking entrance to the allocated slot using an optimized navigation path, reducing unnecessary movement.
- **Real-Time Occupancy Monitoring:** The system continuously monitors parking slot occupancy and availability, ensuring that the displayed information remains updated.
- **Vehicle Entry & Exit Management:** It efficiently manages vehicle entry, parking, and exit operations, while maintaining the current status of each parking space.
- **Automatic Slot Status Update:** When a vehicle exits, the system automatically changes the slot status from Occupied → Available, helping reduce travel time, internal traffic congestion, and unnecessary fuel consumption.



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MODULE 2: OUTPUT



31/8/2026, 11:17:26 am

Parking Map & Navigation

Available Occupied Reserved / Blocked EV VIP Disabled

Zone A

A1 Occupied	A2 Available	A3 Occupied	A4 Occupied	A5 Available	A6 Occupied
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Zone B

B1 Available	B2 Occupied	B3 Occupied	B4 Occupied	B5 Available	B6 Occupied
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Zone C

C1 Available	C2 Occupied	C3 Available	C4 Occupied	C5 Occupied	C6 Available
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Zone D

D1 Occupied	D2 Available	D3 Available	D4 Occupied	D5 Available	D6 Occupied
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Slot B5

Status	Available
Type	Regular
Vehicle	—
Distance from Entrance	150 m
Congestion	68%
Current Score	33/100

Navigation (Graph + Dijkstra's Algorithm)

Select an occupied slot to compute the shortest route from Entrance.

B3 (TN60GH8319)

Entrance → Junction-B → B3

Total Distance	183 m
Estimated Travel Time	2.6 min
Congestion Level	High



MODULE 3: BILLING, ANALYTICS, AND SMART MONITORING



- Automated Billing: Calculate parking charges automatically based on the total parking duration, reducing manual calculation errors.
- Digital Bills & Receipts: Generate digital payment bills and receipts for users after completing their parking session.
- Parking History Management: Maintain detailed parking history and transaction records, including vehicle, slot, duration, and payment information.
- Parking Analytics: Analyze parking usage, occupancy rates, and peak parking hours to understand parking patterns and improve management.
- Smart Monitoring & Alerts: Monitor parking activities in real time, identify unauthorized parking, and generate reports to help administrators optimize parking-space utilization.

MODULE 3: BILLING, ANALYTICS, AND SMART MONITORING

Bill Receipt

Parking ID	P1006
Vehicle Number	TN60GH8319
Slot	B3
Entry Time	31/8/2026, 9:26:36 am
Exit Time	31/8/2026, 11:19:33 am
Duration	1.88 hr
Rate	₹20/hour
Total Amount	₹40
Print Bill	

Algorithm Comparison

31/8/2026, 11:21:42 am

Compare Parking Allocation Algorithms

[Run Comparison](#)

Simulates 12 incoming vehicles against a snapshot of the current slot layout for each strategy. Values are generated live from the simulation, not hard-coded.

ALGORITHM	SEARCH TIME (MS)	AVG WAITING/TRAVEL TIME (MIN)	SPACE UTILIZATION
First Available	2.52	2.05	100%
Nearest Slot	4.32	2.05	100%
Priority Queue	7.24	2.05	100%
Proposed Algorithm Best	7.24	2.05	100%

CONCLUSION

- The Smart Parking Management System provides an efficient and automated solution for managing modern parking facilities.
- The system uses graphs, priority queues, hash tables, queues, and linked lists to manage parking operations effectively.
- Smart slot allocation and optimized navigation help reduce vehicle waiting time and parking congestion.
- Billing, parking history, and real-time monitoring make parking operations more organized and convenient.
- Predictive analytics helps estimate parking demand and improve the overall utilization of available parking spaces.



REFERENCES



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THANK YOU